Exercise No. 2 ADVANCED ENCRYPTION STANDARD

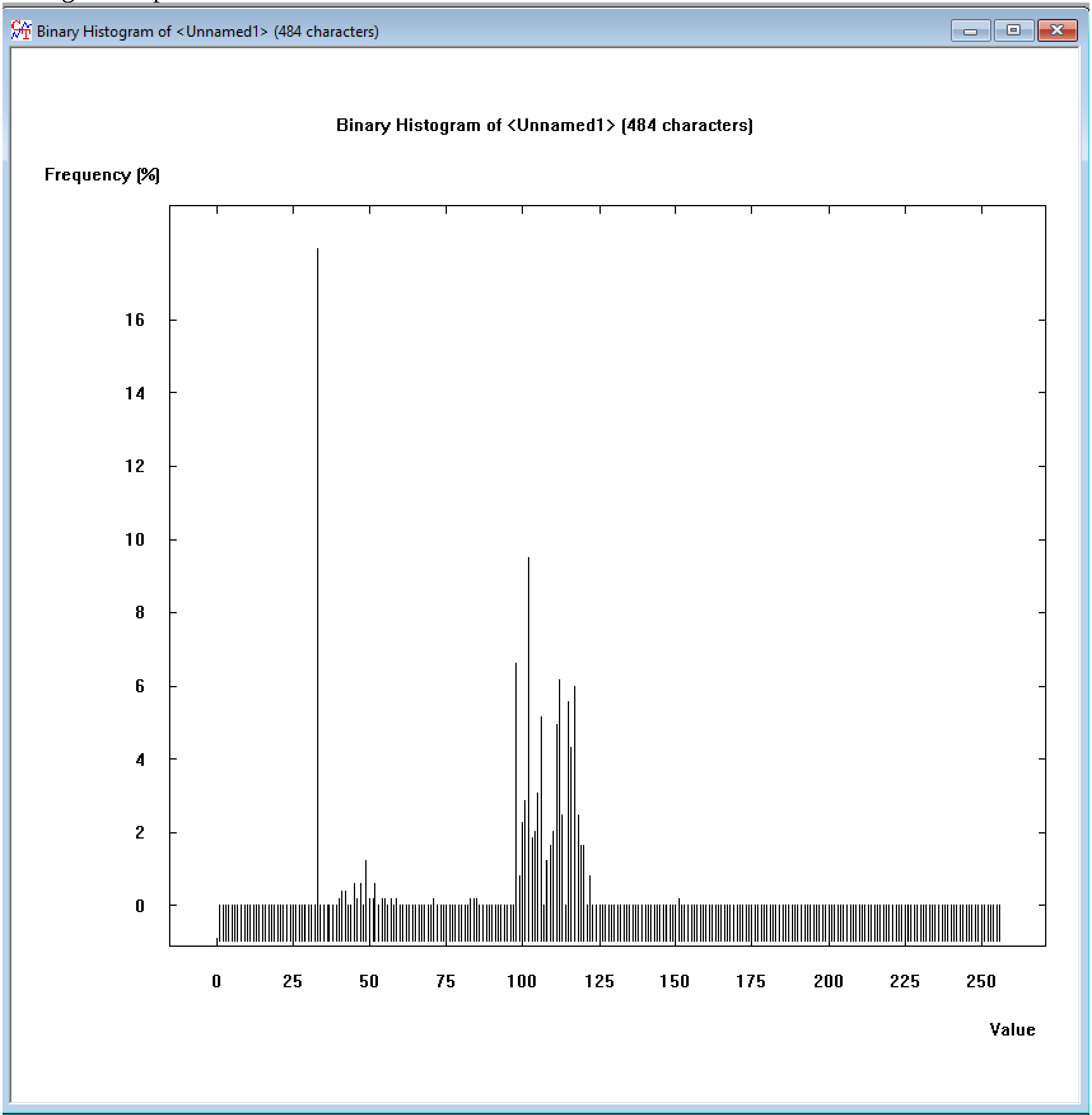
Adam Cherek, s207250

3.1

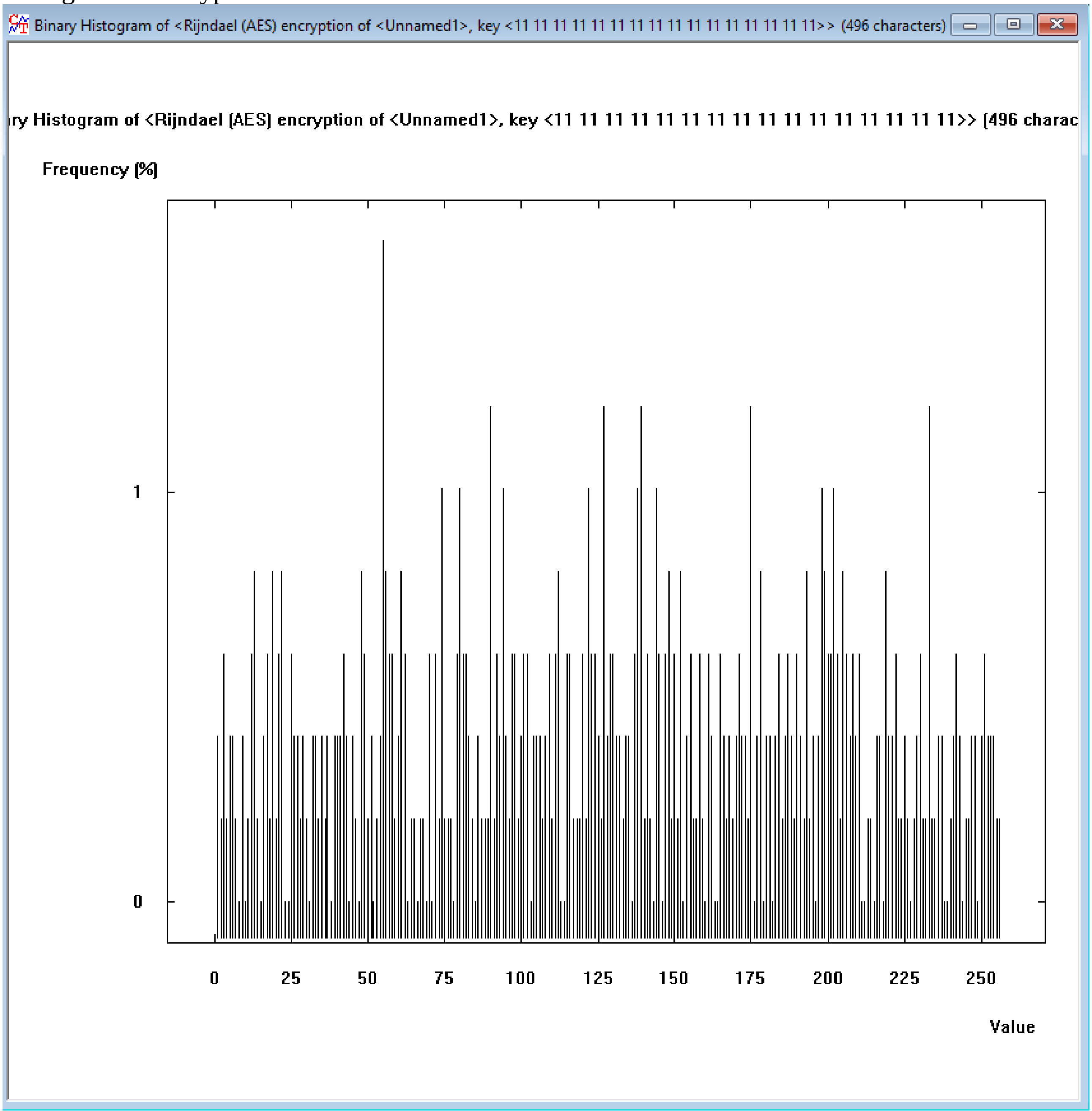
Plain text:
Store leftover pancakes in an airtight container in the fridge for about a week. Refrain from adding toppings (such as syrup) until right before you serve them so the pancakes don't get soggy. To reheat pancakes, you have a few options: warm a batch in a 350 degrees F oven (covered) for 8 to 10 minutes, microwave individual pancakes in 20 to 30 second bursts with a damp paper towel, or reheat them in a skillet over medium-low heat for 30–60 seconds per side until warmed through.

Key: 11111111111111111111111111111111

Histogram of plain text:

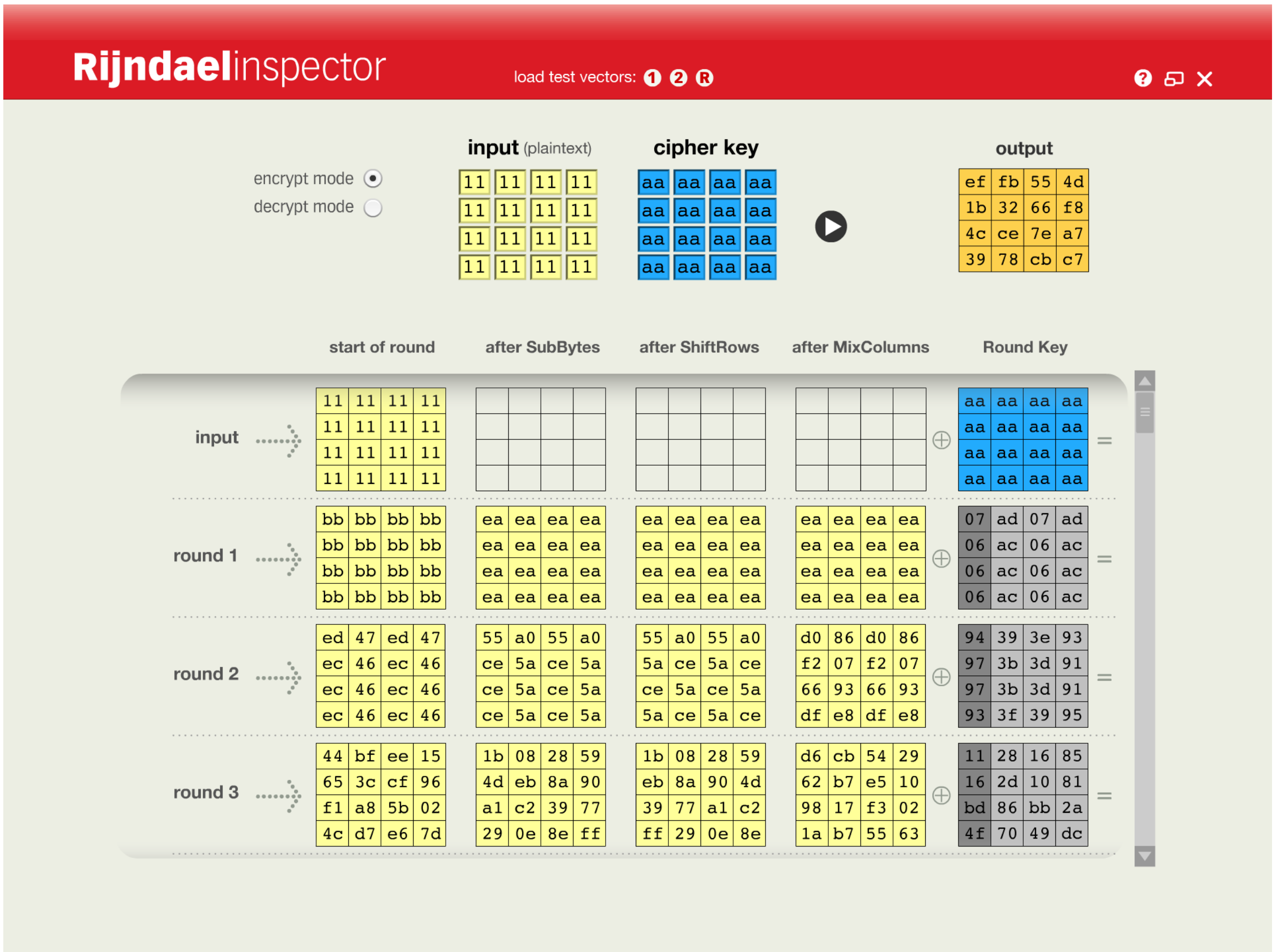


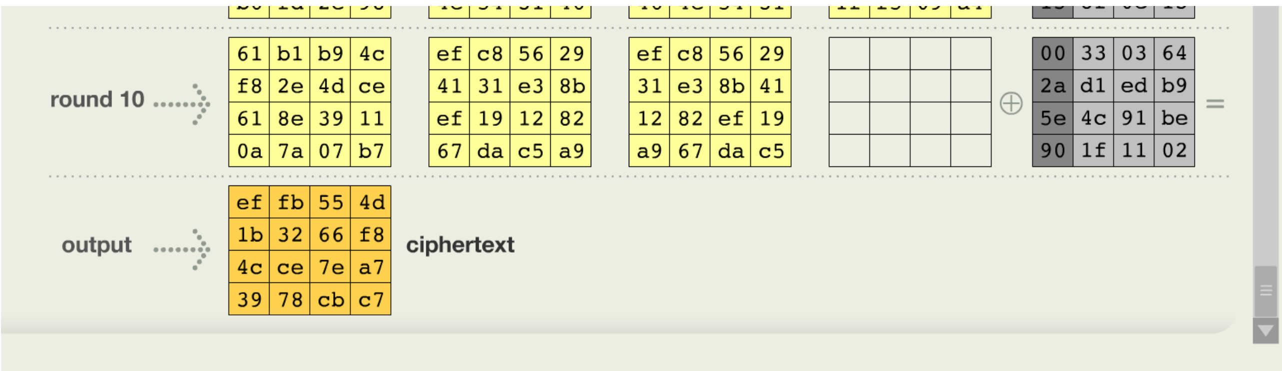
Histogram of encryption:



Conclusions:
AES exhibits excellent "confusion" and "diffusion." The uniform histogram proves that the algorithm successfully destroys any statistical patterns present in the plaintext, rendering frequency analysis attacks completely useless. Decrypting the text with the same key will perfectly restore the original uneven histogram.

3.2





After how many rounds, the outcome is similar to random noise?:
Looks like after 2-3.

Why do we continue up to 10 times?:
To ensure complete security. While 3 rounds might look visually random to a human, mathematical correlations still exist. Ten rounds guarantee that every single bit of the output ciphertext is a complex, non-linear function of every single bit of the plaintext and the key, making cryptanalysis computationally infeasible.

3.3

Plain text:

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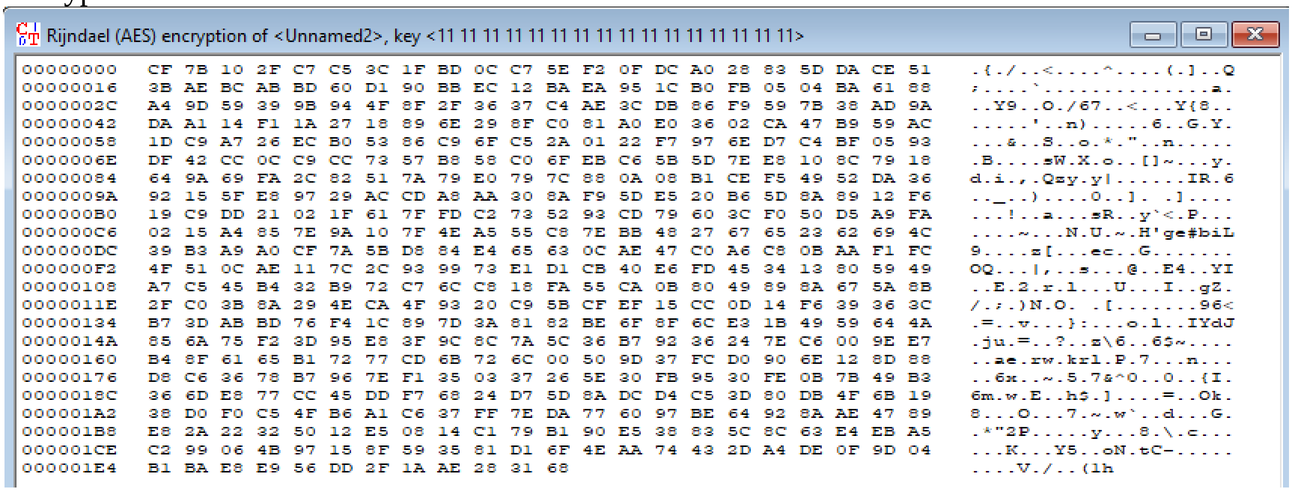
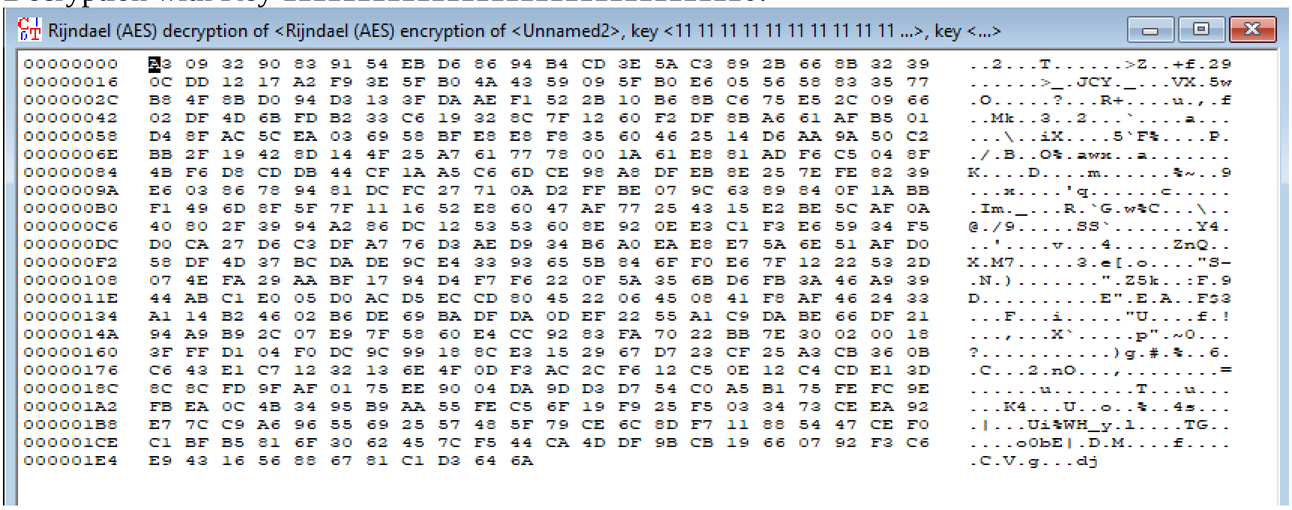
Key: 11111111111111111111111111111111

Encryption:



Decryption with Key 11111111111111111111111111111110:



Results:
Decrypted text is completely unreadable. Observed effect – the Avalanche Effect (specifically, the Strict Avalanche Criterion). In a well-designed cipher like AES, a change in just 1 bit of the key (or plaintext) results in approximately a 50% change in the resulting output bits. Because the decryption key schedule diverges entirely from the correct key schedule after that 1-bit flip, the algorithm outputs random noise instead of the original text.

3.4

Plain text:

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Key: 1111111111111111111111111111111

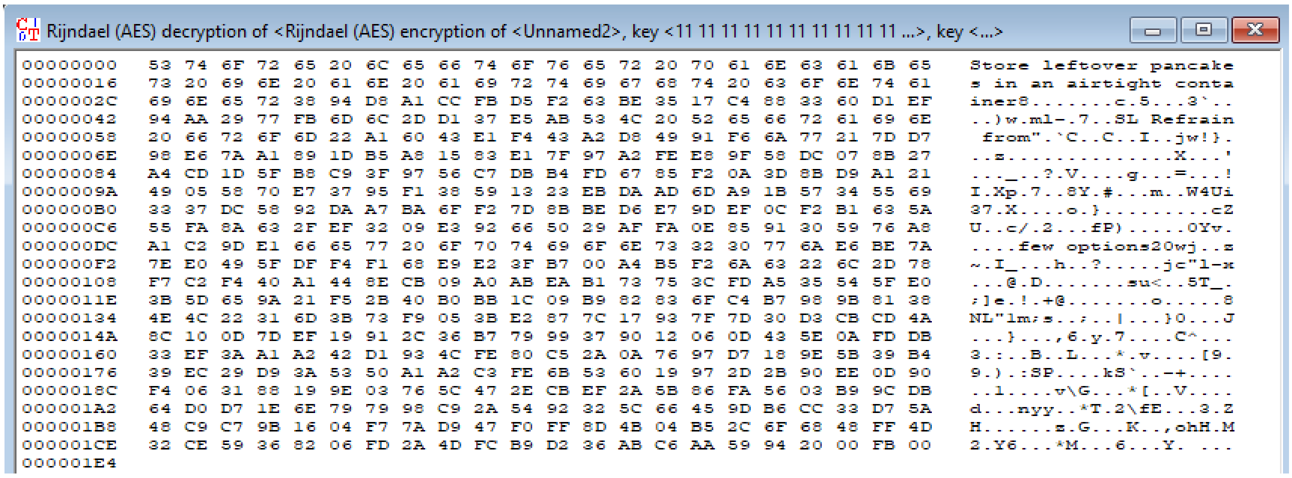
Encryption:

```
Rijndael (AES) encryption of <Unnamed2>, key <11 11 11 11 11 11 11 11 11 11 11 11 11 11 11 11>

00000000  CF 7B 10 2F C7 C5 3C 1F BD 0C C7 5E F2 0F DC A0 28 83 5D DA CE 51  .{./..<....^....(.]..Q
00000016  3B AE BC AB BD 60 D1 90 BB EC 12 BA EA 95 1C B0 FB 05 04 BA 61 88  ;....`..............a.
0000002C  A4 9D 59 39 9B 94 4F 8F 2F 36 37 C4 AE 3C DB 86 F9 59 7B 38 AD 9A  ..Y9..O./67..<...Y{8..
00000042  DA A1 14 F1 1A 27 18 89 6E 29 8F C0 81 A0 E0 36 02 CA 47 B9 59 AC  .....'..n).....6..G.Y.
00000058  1D C9 A7 26 EC B0 53 86 C9 6F C5 2A 01 22 F7 97 6E D7 C4 BF 05 93  ...&..S..o.*."..n.....
0000006E  DF 42 CC 0C C9 CC 73 57 B8 58 C0 6F EB C6 5B 5D 7E E8 10 8C 79 18  .B....sW.X.o..[]~...y.
00000084  64 9A 69 FA 2C 82 51 7A 79 E0 79 7C 88 0A 08 B1 CE F5 49 52 DA 36  d.i.,.Qzy.y|......IR.6
0000009A  92 15 5F E8 97 29 AC CD A8 AA 30 8A F9 5D E5 20 B6 5D 8A 89 12 F6  .._..).....0..]. .]....
000000B0  19 C9 DD 21 02 1F 61 7F FD C2 73 52 93 CD 79 60 3C F0 50 D5 A9 FA  ...!..a...sR..y`<.P...
000000C6  02 15 A4 85 7E 9A 10 7F 4E A5 55 C8 7E BB 48 27 67 65 23 62 69 4C  ....~...N.U.~.H'ge#biL
000000DC  39 B3 A9 A0 CF 7A 5B D8 84 E4 65 63 0C AE 47 C0 A6 C8 0B AA F1 FC  9....z[...ec..G.......
000000F2  4F 51 0C AE 11 7C 2C 93 99 73 E1 D1 CB 40 E6 FD 45 34 13 80 59 49  OQ...|,..s...@..E4..YI
00000108  A7 C5 45 B4 32 B9 72 C7 6C C8 18 FA 55 CA 0B 80 49 89 8A 67 5A 8B  ..E.2.r.l...U...I..gZ.
0000011E  2F C0 3B 8A 29 4E CA 4F 93 20 C9 5B CF EF 15 CC 0D 14 F6 39 36 3C  /.;.)N.O. .[.......96<
00000134  B7 3D AB BD 76 F4 1C 89 7D 3A 81 82 BE 6F 8F 6C E3 1B 49 59 64 4A  .=..v...}:...o.l..IYdJ
0000014A  85 6A 75 F2 3D 95 E8 3F 9C 8C 7A 5C 36 B7 92 36 24 7E C6 00 9E E7  .ju.=..?..z\6..6$~....
00000160  B4 8F 61 65 B1 72 77 CD 6B 72 6C 00 50 9D 37 FC D0 90 6E 12 8D 88  ..ae.rw.krl.P.7...n...
00000176  D8 C6 36 78 B7 96 7E F1 35 03 37 26 5E 30 FB 95 30 FE 0B 7B 49 B3  ..6x..~.5.7&^0..0..{I.
0000018C  36 6D E8 77 CC 45 DD F7 68 24 D7 5D 8A DC D4 C5 3D 80 DB 4F 6B 19  6m.w.E..h$.].....=..Ok.
000001A2  38 D0 F0 C5 4F B6 A1 C6 37 FF 7E DA 77 60 97 BE 64 92 8A AE 47 89  8...O...7.~.w`..d...G.
000001B8  E8 2A 22 32 50 12 E5 08 14 C1 79 B1 90 E5 38 83 5C 8C 63 E4 EB A5  .*"2P.....y...8.\.c...
000001CE  C2 99 06 4B 97 15 8F 59 35 81 D1 6F 4E AA 74 43 2D A4 DE 0F 9D 04  ...K...Y5..oN.tC-.....
000001E4  B1 BA E8 E9 56 DD 2F 1A AE 28 31 68                                ....V./..(1h
```

Decryption of slightly changed text:

```
Rijndael (AES) decryption of <Rijndael (AES) encryption of <Unnamed2>, key <11 11 11 11 11 11 11 11 11 11 ...>, key <...>

00000000  53 74 6F 72 65 20 6C 65 66 74 6F 76 65 72 20 70 61 6E 63 61 6B 65  Store leftover pancake
00000016  73 20 69 6E 20 61 6E 20 61 69 72 74 69 67 68 74 20 63 6F 6E 74 61  s in an airtight conta
0000002C  69 6E 65 72 20 69 6E 20 74 68 65 20 66 72 69 64 67 65 20 66 6F 72  iner in the fridge for
00000042  20 61 62 6F 75 74 20 61 20 77 65 65 6B 2E 20 52 65 66 72 61 69 6E   about a week. Refrain
00000058  20 66 72 6F 6D 20 61 64 64 69 6E 67 20 74 6F 70 70 69 6E 67 73 20   from adding toppings 
0000006E  28 73 7B 3D 67 0F DB 51 02 0F 35 A5 41 C2 CC 60 95 CF 85 1A 09 BB  (s{=g..Q..5.A..`......
00000084  03 51 B2 50 E8 46 DB E5 42 E0 DC BC 01 D6 0F 75 20 73 65 72 76 65  .Q.P.F..B......u serve
0000009A  20 74 68 65 6D 20 73 6F 20 74 68 65 20 70 61 6E 63 61 6B 65 73 20   them so the pancakes 
000000B0  8D 52 DB BA 13 99 83 64 78 A8 39 DC 41 24 6D 0A 20 54 6F 20 72 6B  .R.....dx.9.A$m. To rk
000000C6  68 65 61 74 20 70 61 6E 63 61 6B 65 73 2C 20 79 6F 75 20 68 61 76  heat pancakes, you hav
000000DC  65 20 61 20 66 65 77 20 6F 70 74 69 6F 6E 73 3A 20 77 61 72 27 E1  e a few options: war'.
000000F2  FA 6E 32 B9 31 2B F3 C3 E4 42 61 25 88 FD 35 30 20 64 65 67 72 65  .n2.1+...Ba%..50 degre
00000108  68 F3 20 46 20 6F 76 65 6E 20 28 63 6F 76 65 72 65 64 29 20 66 6F  h. F oven (covered) fo
0000011E  72 20 38 20 74 6F 20 31 30 20 6D 69 6E 75 74 65 73 2C 20 6D 69 63  r 8 to 10 minutes, mic
00000134  72 6F 77 61 76 65 20 69 6E 64 69 76 69 64 75 61 6C 20 70 61 6E 63  rowave individual panc
0000014A  61 6B 65 73 20 69 6E 20 32 30 20 74 6F 20 33 30 20 73 65 63 6F 6E  akes in 20 to 30 secon
00000160  0D D2 5D 1E FD 70 A9 02 A0 33 D3 D6 55 D4 CD E0 E2 FD 1E 20 20 70  ..]..p...3..U......  p
00000176  67 AC 15 72 20 74 6F 77 65 6C 2C 20 6F 72 20 72 65 68 65 61 74 20  g..r towel, or reheat 
0000018C  74 68 65 6D 20 69 6E 20 61 20 73 6B 69 6C 6C 65 74 20 6F 76 65 72  them in a skillet over
000001A2  20 6D 65 64 69 75 6D 2D 6C 6F 77 20 68 65 61 74 20 66 6F 72 20 33   medium-low heat for 3
000001B8  30 96 36 30 20 73 65 63 6F 6E 64 73 20 70 65 72 20 73 69 64 65 20  0.60 seconds per side 
000001CE  75 6E 74 69 6C 20 77 61 72 6D 65 64 20 74 68 72 6F 75 67 68 2E 20  until warmed through. 
000001E4
```

Decryption of massively changed text:

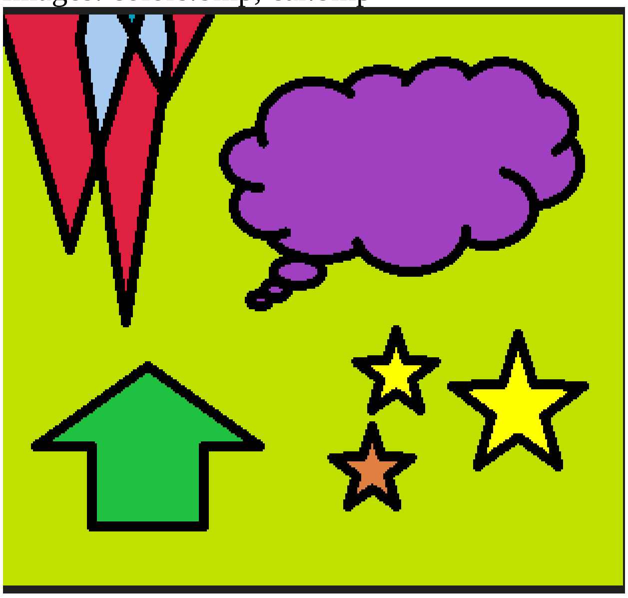


```
Rijndael (AES) decryption of <Rijndael (AES) encryption of <Unnamed2>, key <11 11 11 11 11 11 11 11 11 11 ...>, key <...>
00000000  53 74 6F 72 65 20 6C 65 66 74 6F 76 65 72 20 70 61 6E 63 61 6B 65  Store leftover pancake
00000016  73 20 69 6E 20 61 6E 20 61 69 72 74 69 67 68 74 20 63 6F 6E 74 61  s in an airtight conta
0000002C  69 6E 65 72 38 94 D8 A1 CC FB D5 F2 63 BE 35 17 C4 88 33 60 D1 EF  iner8.......c.5...3`..
00000042  94 AA 29 77 FB 6D 6C 2D D1 37 E5 AB 53 4C 20 52 65 66 72 61 69 6E  ..)w.ml-.7..SL Refrain
00000058  20 66 72 6F 6D 22 A1 60 43 E1 F4 43 A2 D8 49 91 F6 6A 77 21 7D D7   from".`C..C..I..jw!}.
0000006E  98 E6 7A A1 89 1D B5 A8 15 83 E1 7F 97 A2 FE E8 9F 58 DC 07 8B 27  ..z..............X...'
00000084  A4 CD 1D 5F B8 C9 3F 97 56 C7 DB B4 FD 67 85 F2 0A 3D 8B D9 A1 21  ..._..?.V....g...=...!
0000009A  49 05 58 70 E7 37 95 F1 38 59 13 23 EB DA AD 6D A9 1B 57 34 55 69  I.Xp.7..8Y.#...m..W4Ui
000000B0  33 37 DC 58 92 DA A7 BA 6F F2 7D 8B BE D6 E7 9D EF 0C F2 B1 63 5A  37.X....o.}.........cZ
000000C6  55 FA 8A 63 2F EF 32 09 E3 92 66 50 29 AF FA 0E 85 91 30 59 76 A8  U..c/.2...fP).....0Yv.
000000DC  A1 C2 9D E1 66 65 77 20 6F 70 74 69 6F 6E 73 32 30 77 6A E6 BE 7A  ....few options20wj..z
000000F2  7E E0 49 5F DF F4 F1 68 E9 E2 3F B7 00 A4 B5 F2 6A 63 22 6C 2D 78  ~.I_...h..?.....jc"l-x
00000108  F7 C2 F4 40 A1 44 8E CB 09 A0 AB EA B1 73 75 3C FD A5 35 54 5F E0  ...@.D.......su<..5T_.
0000011E  3B 5D 65 9A 21 F5 2B 40 B0 BB 1C 09 B9 82 83 6F C4 B7 98 9B 81 38  ;]e.!.+@.......o.....8
00000134  4E 4C 22 31 6D 3B 73 F9 05 3B E2 87 7C 17 93 7F 7D 30 D3 CB CD 4A  NL"1m;s..;..|...}0...J
0000014A  8C 10 0D 7D EF 19 91 2C 36 B7 79 99 37 90 12 06 0D 43 5E 0A FD DB  ...}...,6.y.7....C^...
00000160  33 EF 3A A1 A2 42 D1 93 4C FE 80 C5 2A 0A 76 97 D7 18 9E 5B 39 B4  3.:..B..L...*.v....[9.
00000176  39 EC 29 D9 3A 53 50 A1 A2 C3 FE 6B 53 60 19 97 2D 2B 90 EE 0D 90  9.).:SP....kS`..-+....
0000018C  F4 06 31 88 19 9E 03 76 5C 47 2E CB EF 2A 5B 86 FA 56 03 B9 9C DB  ..1....v\G...*[..V....
000001A2  64 D0 D7 1E 6E 79 79 98 C9 2A 54 92 32 5C 66 45 9D B6 CC 33 D7 5A  d...nyy..*T.2\fE...3.Z
000001B8  48 C9 C7 9B 16 04 F7 7A D9 47 F0 FF 8D 4B 04 B5 2C 6F 68 48 FF 4D  H......z.G...K..,ohH.M
000001CE  32 CE 59 36 82 06 FD 2A 4D FC B9 D2 36 AB C6 AA 59 94 20 00 FB 00  2.Y6...*M...6...Y. ...
000001E4
```

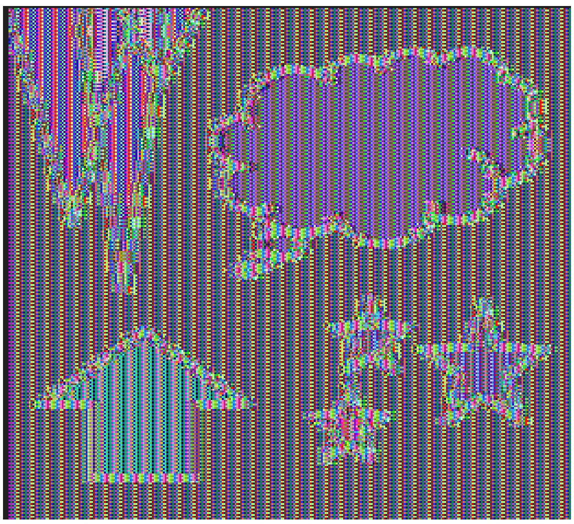
Results:
AES operates on 16-byte blocks. Any error within a ciphertext block ruins the entire corresponding plaintext block upon decryption. Introducing a whole series of changes will permanently corrupt every 16-byte block touched by those changes.

3.5

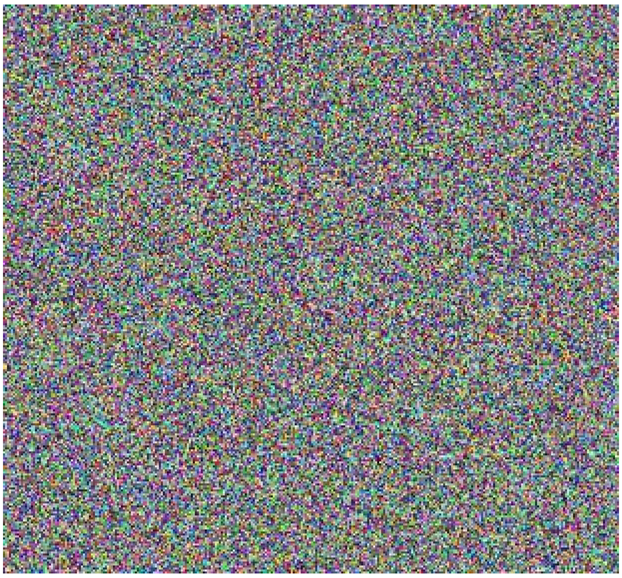
Images: colors.bmp, car.bmp

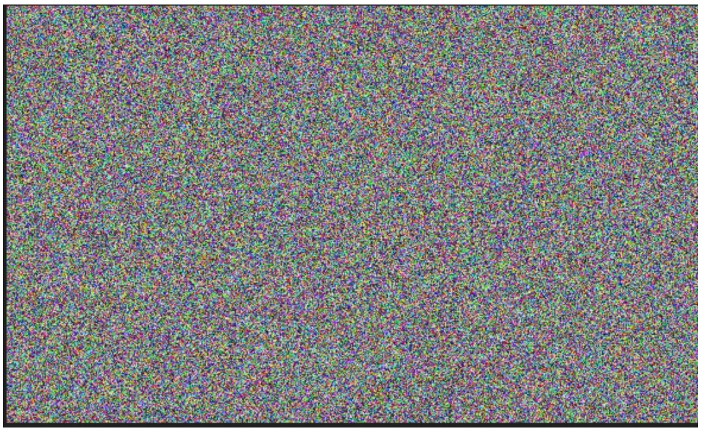



Encrypted ECB:

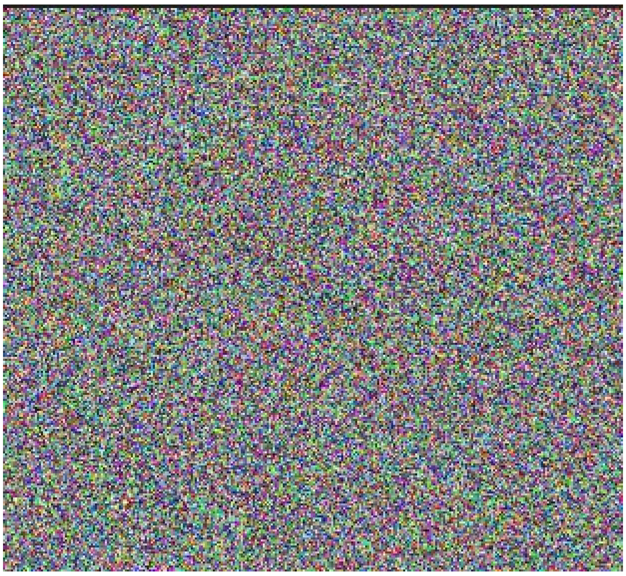


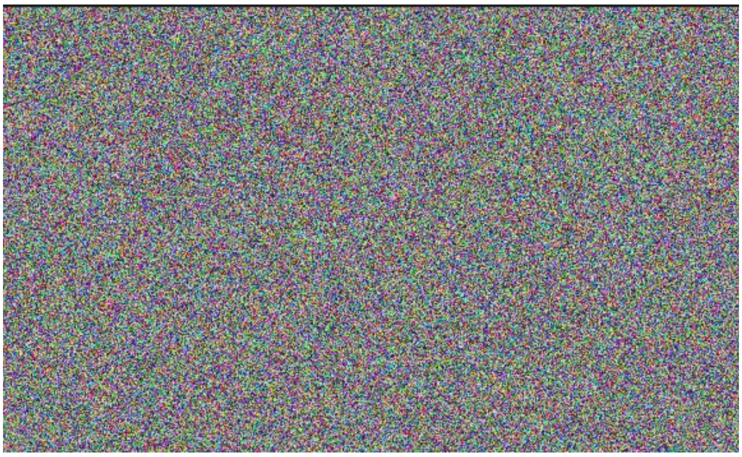

Encrypted CBC:





Encrypted PCBC:





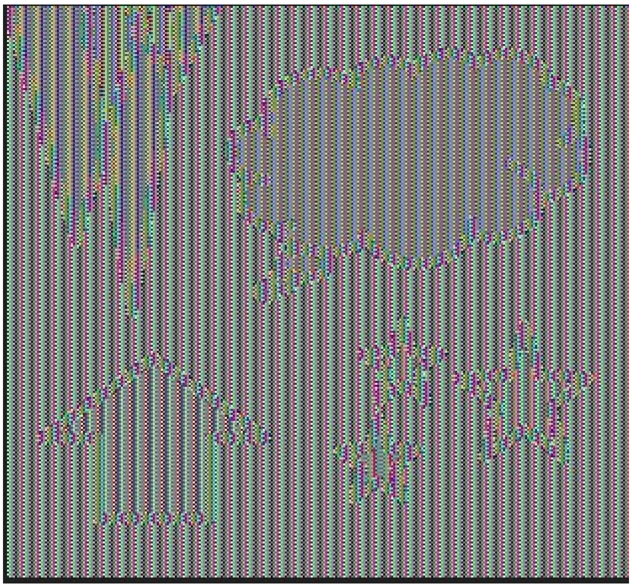
Comparison:

- ECB: The encrypted image still displays the clear visual outline, shapes, and color groupings of the original car or colors. Identical plaintext blocks of color are encrypted into identical ciphertext blocks.
- CBC: The encrypted image looks like pure static or TV snow. No visual patterns from the original image remain.
- PCBC: Similar to CBC, the image looks like completely random static noise.

Conclusions:
ECB mode is highly insecure for data with repeating patterns (like image headers or solid color blocks) because it does not hide the underlying data structure. CBC and PCBC modes XOR the current block with the previous ciphertext block, effectively hiding all repetitive patterns.

3.6

Decrypted PCBC with error:



Results:

- ECB: The bottom-left corner of the image has a small block of scrambled, colorful static. The entire rest of the image is perfect.
- CBC: The bottom-left corner is scrambled, and the block immediately following it has corrupted pixels. The rest of the image recovers and displays perfectly.
- PCBC: The bottom-left corner is scrambled, and the entire rest of the image is somehow readable but changed.

Conclusions:
PCBC mode has an infinite error propagation property. An error in one block cascades forward and corrupts every single block that comes after it during decryption.

3.7

Time for 13 bits key: 10 seconds

Time for 12 bits key: 1 hour

Time for 11 bits key: 10,5 days

Results:
It takes multiple orders of magnitude more time to brute force AES than the old DES.

3.8

Message (decrypted in 3.7):
Lollipops are available in a number of colors and flavors, particularly fruit flavors. With numerous companies producing lollipops, the candy now comes in dozens of flavors and many different shapes. They range from small ones which can be bought by the hundred and are often given away for free at banks, barbershops, etc., to very large ones made out of candy canes twisted into a circle. Some lollipops contain fillings, such as bubble gum or soft candy. Some novelty lollipops have more unusual items, such as mealworm larvae, embedded in the candy. Other novelty lollipops have

non-edible centers, such a flashing light, embedded within the candy; there is also a trend[where?] of lollipops with sticks attached to a motorized device that makes the entire lollipop spin around in one's mouth.
Some lollipops have been marketed for use as diet aids, although their effectiveness is untested, and anecdotal cases of weight loss may be due to the power of suggestion. Flavored lollipops made with children's medicine have also been created in order to effectively give kids medicine without fuss. Drugmakers have also developed lollipops containing fentanyl, a powerful analgesic.
In the Nordic countries, Germany, and the Netherlands, lollipops may be flavored with salmiak.

Brute Force decryption time for 128 bit key: 8.9e+24 years

Brute Force decryption time for 192 bit key: 1.8e+44 years

Brute Force decryption time for 256 bit key: 3.9e+63 years

Results:

- Attacking AES-192 takes 2^64 times longer than attacking AES-128.
- Attacking AES-256 takes 2^128 times longer than attacking AES-128.

3.9

Advantages: -

- Highly Secure: Resistant to all known practical attacks (differential and linear cryptanalysis).
- Efficient and Fast: Performs extremely well in both software and hardware across various platforms.
- Flexible Key Sizes: Offers 128, 192, and 256-bit lengths depending on security needs.
- Standardized: Globally accepted, thoroughly peer-reviewed, and standardized by NIST.

Disadvantages:

- Symmetric Key Issue: Both parties must safely exchange the key beforehand; if the key is intercepted, the system falls.
- Implementation Risks: If programmed poorly, AES is vulnerable to side-channel attacks (like timing or power analysis attacks).
- Mode Dependencies: Relying on weak modes of operation (like ECB) undermines the entire security of the algorithm.